

Quantum Optimization for EV Charging Using QAOA

1. Introduction

Efficient scheduling of electric vehicle (EV) charging is an increasingly important problem as EV adoption grows and electrical grids face higher demand variability. The objective is to allocate charging power across multiple vehicles and time slots while minimizing inefficiencies such as energy loss and peak load stress.

This project models the EV charging scheduling problem as an optimization problem and solves it using both classical and quantum-inspired approaches in a similar fashion as Sturm, A. (2023). *Theory and Implementation of the Quantum Approximate Optimization Algorithm*. In particular, the Quantum Approximate Optimization Algorithm (QAOA) is used to approximate optimal solutions after converting the problem into a Quadratic Unconstrained Binary Optimization (QUBO) form.

The workflow includes:

- Modeling the charging system
- Converting constraints into QUBO form
- Solving using classical and QAOA-based methods
- Evaluating performance under realistic conditions (noise, efficiency, losses)

2. Problem Formulation

2.1 System Model

The system consists of:

- Multiple EVs (*car_green* and *car_orange*)
- Discrete time slots
- A shared charging unit

Each vehicle is defined by:

- Available time slots
- Required energy demand

A Car class is used to capture these properties. And also the charging unit aggregates all vehicles into one optimization problem.

2.2 Optimization Objective

The goal is to:

- Deliver required energy to each car
- Minimize power inefficiencies
- Avoid excessive peak demand

This results in a quadratic objective function that penalizes:

- High power usage
- Simultaneous charging (cross terms)
- Energy loss (modeled later)

3. QUBO Conversion

To enable quantum optimization, the problem is converted into QUBO form, which has the structure:

$$\text{Minimize } x^T Q x$$

3.1 Conversion Steps

Two key transformations are applied:

1. LinearEqualityToPenalty
 - Converts constraints into penalty terms
 - Adds a penalty factor $\rho = 5.1$
2. IntegerToBinary
 - Converts integer variables into binary variables
 - Required for quantum algorithms

This transformation ensures the problem becomes:

- Unconstrained
- Fully expressed in binary variables

4. QAOA Implementation

4.1 Baseline (Unoptimized QAOA)

A small-scale version of the problem is used to avoid computational limitations, but the other copy of the QAOA demonstrates parallelism between the objective scores, but with a long (2 hours) runtime. The QUBO is converted into an Ising Hamiltonian, which is then solved using a classical eigensolver as a stand-in for quantum execution.

This provides:

- A reference solution
- Validation of correctness

4.2 Parameter Optimization ($p = 1$)

QAOA introduces two parameters:

- β : mixing angle
- γ : cost angle

A classical optimizer (COBYLA) is used to minimize the QAOA cost function.

Key points:

- The cost function evaluates expectation values of the Hamiltonian
- The classical optimum is used as a benchmark
- Optimization improves solution quality compared to random parameters

4.3 Increased Circuit Depth ($p = 2$)

Increasing QAOA depth improves functionality, but too much will increase the objective score.

For depth $p = 2$:

- More parameters are introduced
- The solution space becomes richer
- Better approximations of the optimal solution are possible

A custom cost function simulates this deeper circuit using layered cosine approximations.

5. Noisy Quantum Simulation

Real quantum devices are noisy, so a simulation using:

- AerSimulator
- FakeVigoV2 backend

is performed.

5.1 Circuit Construction

- Hadamard gates initialize superposition
- Alternating cost and mixing layers simulate QAOA

- Parameters from optimization are applied

5.2 Observations

- Noise degrades solution quality
- Demonstrates the gap between ideal and real quantum hardware
- Highlights the importance of error mitigation

6. Physical Power and Efficiency Analysis

Beyond optimization, the project incorporates real-world electrical considerations.

6.1 Efficiency Model

Charging efficiency varies with power:

- 1 kW $\rightarrow$ 95%
- 6 kW $\rightarrow$ 90%

6.2 Loss Model

Losses are modeled using:

$$\text{Loss} \propto I^2 R \sim P^2$$

Implemented as:

- $\alpha = 0.01$
- Loss increases quadratically with power

6.3 Energy Metrics

The following are computed:

- Energy delivered to users
- Energy drawn from grid
- Total losses
- Overall efficiency

This provides a more realistic evaluation of scheduling performance.

7. Baseline Comparison

A baseline scenario is defined:

- Both cars charge at full power (6 kW) at all times

Results Comparison

Metrics include:

- Total grid energy
- Delivered energy
- Losses
- Efficiency

The optimized schedule demonstrates:

- Reduced losses
- Improved efficiency
- Lower stress on the grid

8. Load Factor Analysis

Load factor is a key grid performance metric:

$$\text{Load Factor} = \frac{\text{Average Load}}{\text{Peak Load}}$$

Interpretation

- High load factor → efficient usage
- Low load factor → high peaks, inefficient grid usage

The optimized schedule:

- Reduces peak load
- Improves load factor
- Leads to more stable grid operation

9. Conclusion

This project demonstrates how quantum optimization techniques can be applied to a real-world engineering problem.

Key Contributions

- Formulated EV charging as a QUBO problem
- Applied QAOA for approximate optimization

- Compared classical, quantum-inspired, and noisy simulations
- Integrated realistic electrical models (efficiency, losses)

Key Insights

- QAOA can approximate optimal schedules effectively
- Increasing circuit depth improves solution quality
- Noise remains a major limitation in practical quantum systems
- Optimized charging significantly improves energy efficiency and grid stability

10. Future Work

Potential improvements include:

- Running on real quantum hardware
- Scaling to larger EV fleets
- Incorporating dynamic pricing or renewable energy sources
- Improving parameter optimization strategies